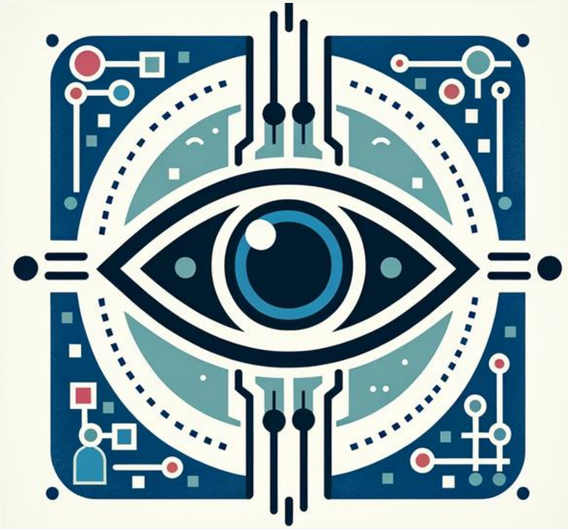


COMPUTER VISION

Project

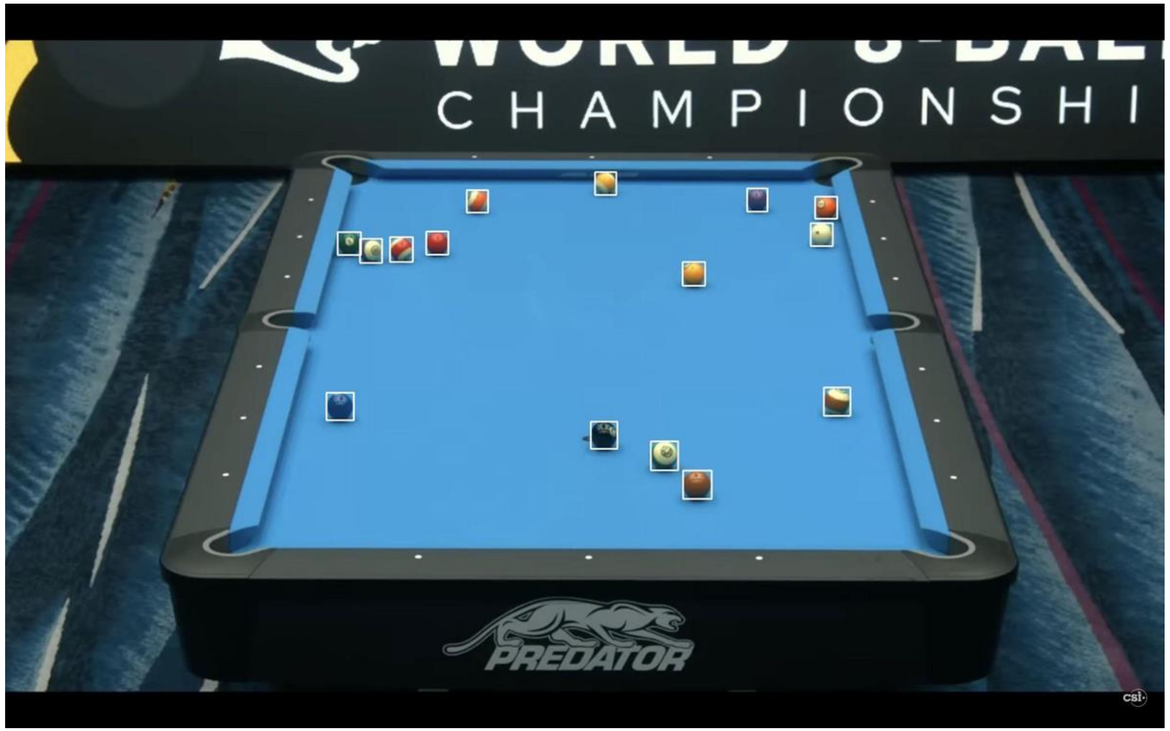
M.EIC+M.IA | FEUP

1
14
10
3

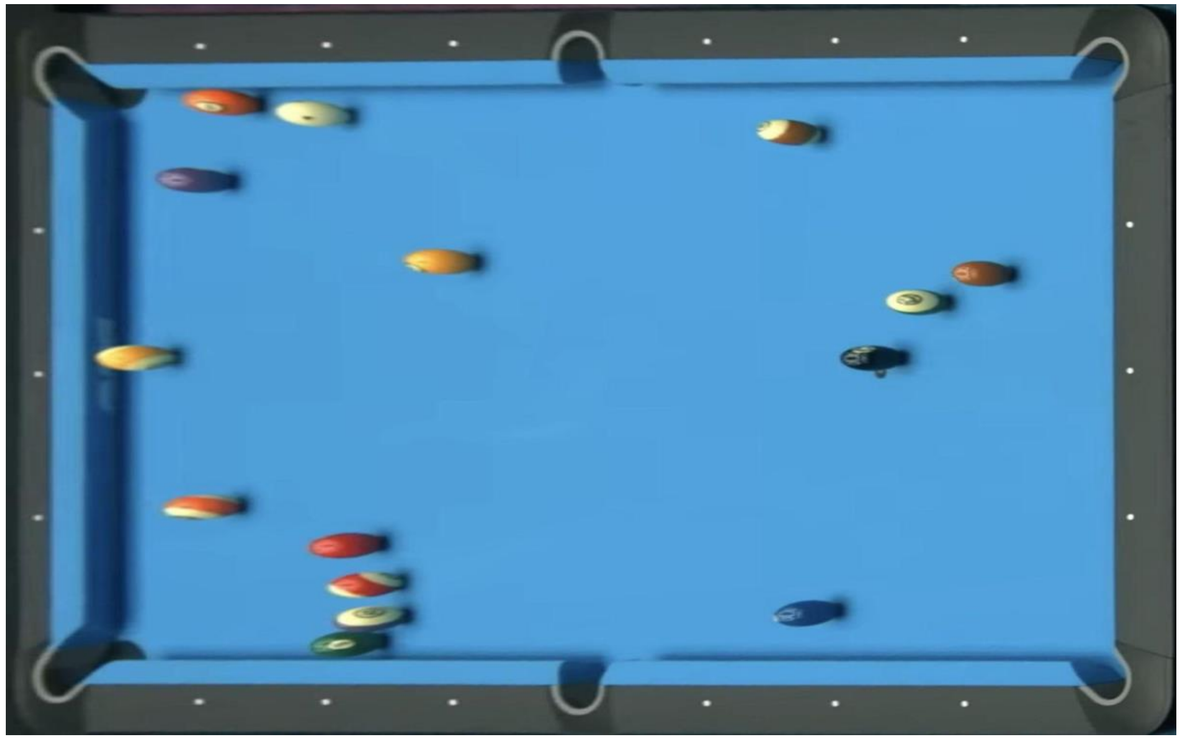
TASK 1

- Input:
  - Image of 8-ball pool table
- Output:
  - Total number of balls on the table
  - Position of the balls on the image (bounding boxes)
  - Number of each ball
  - Top-view of the table

TASK 1



Bounding Box Detection



Top-View of the Table

Dataset: https://universe.roboflow.com/bachelorthesis/8-ball-pool-l530o

TASK 1

- Ball numbers can be obtained based on their colour
- Consider the cue (white) ball to be zero

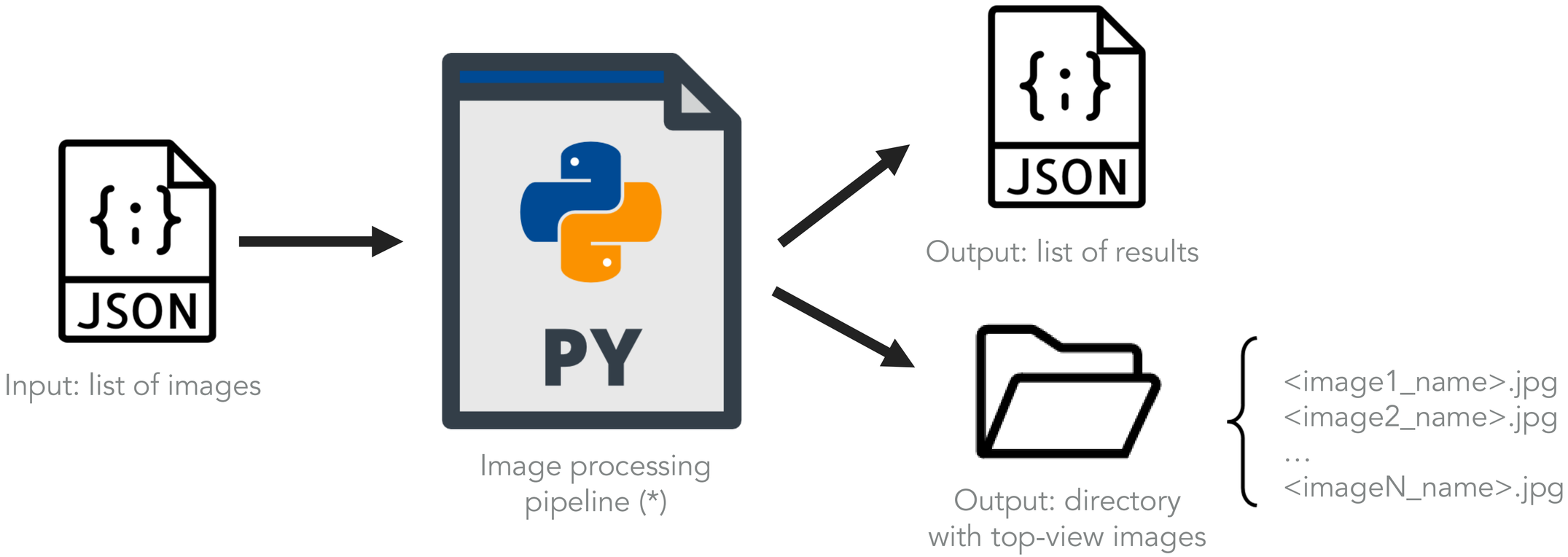
TASK 1

- Dataset:
  - 50 images randomly chosen from a public dataset
  - The results will be tested in 10 **undisclosed** images

- Deliverables:
  - Short report (2 pages max) presenting the **methodology** and some **results**
  - Python script (only one file)

- Deadline: **April, 17** (23:59 AoE)

TASK 1



(*) using only OpenCV and other common libraries, like numpy and matplotlib

TASK 1

- Grading
  - Task 1 accounts for **40%** of the overall project grade
  - Elements being considered: methodology, report and quality of the results

- Important remarks
  - Follow **strictly** the JSON structure for the input and output files
  - It is **okay** to use AI tools while developing your work, but it is **not okay** to use them without acknowledging it
  - All members of the group **are expected to understand** the methodology and the submitted code